Neurons of a Looped Language Model: Neuron-Level Interpretability and Pre-Registered Causal Steering of Ouro-1.4B

Libor Burian

Code: https://github.com/BurnyCoder/looped-transformers-llms-neuron-mechanistic-interpretability

Abstract. Looped language models apply the same layers several times per token, so every MLP neuron fires once per loop. We study the 135,168 SwiGLU neurons of Ouro-1.4B, which loops its 24 layers four times, at the level of single neurons, following Goodfire's raw-neuron methodology.

Atlas. From 8M tokens we build per-loop top-context tables.
- Strongest activations concentrate in early loops (41% at loop 1, 13% at loop 4).
- Same-token activations of a neuron at loops 1 and 4 correlate with median $r = 0.54$, and consecutive loops converge (r: 0.65, 0.84, 0.94).

Pre-registered test. We selected candidates on discovery prompts only and committed a pre-registration before using held-out prompts. Clamping a single neuron at all loops was compared against unsteered generation and against 100 random neurons from the same layer, with Holm correction and fluency and degeneracy guards.
- **All five** pre-registered neurons (phonetics, non-duality, pedagogy, forum answers, electron orbitals) steer 48-token continuations toward their concept: mean Δconcept-score 0.13–0.18, d_z 0.99–1.61, Holm-adjusted Wilcoxon $p \leq 1.1 \times 10^{-15}$, beating 99–100 of 100 random neurons, perplexity ratio ≤ 1.83.
- The effects are concept-specific, confirmed by a blind LLM judge and by teacher-forced probabilities of each neuron's peak-context keywords, stable across three further seed sets, and replicate at identical indices in the reasoning-tuned Ouro-1.4B-Thinking (Wilcoxon criterion fixed before the run; the other replication conditions post hoc).

Follow-ups (post hoc).
- Clamping at any single loop barely steers ($\Delta C \leq 0.043$ versus 0.13–0.17 for all loops), so steering is strongly super-additive across loops, consistent with self-repair.
- Dose-response levels off for four of six neurons, qualitatively consistent with the bound set by the post-MLP RMSNorm.

Negative results.
- Perplexity-based fluency guards accept degenerate repetitions ("Fib Fib Fib…").
- Explanations from a local 9B model transfer across loops as well as within them, so this audit could not detect loop-specific function.

1 Introduction

Looped (recurrent-depth) language models apply one block of layers several times in sequence, which lets them spend more computation per token without adding parameters [1], [2], [3]. Ouro-1.4B [2] loops its entire 24-layer stack four times. Its model card reports performance matching 3–4B-parameter standard transformers [4]. Mechanistic work on looped models has so far studied the residual stream:
- recurrent blocks converge to cyclic fixed points and repeat the "stages of inference" of feedforward models [5];
- residual-direction steering of Ouro-1.4B-Thinking works best at the last loop, and earlier injections are "rotated away" [6];
- per-execution-step refusal directions in Ouro-2.6B align from the second pass onward (a follow-up to a study of Nanbeige4.2-3B) [7].

No published study we are aware of works at the level of **individual neurons** in a looped model.

Neurons are a natural unit for looped models for two reasons. First, a recent essay from Goodfire argues that most MLP neurons of modern models are already interpretable without a learned decomposition [8]. Second, weight sharing creates a test that has no analogue in a feedforward network: the same neuron, with the same weights, is evaluated at every loop. It may do the same job each time, or it may be reused for different jobs at different depths.

We ask three questions:
1. What do Ouro's neurons respond to, and at which loop?
2. Does a neuron keep its function across loops?
3. Can clamping a **single** neuron change what the model writes, with statistical significance against pre-registered controls, including random neurons from the same layer clamped the same way?

Contributions.
- A neuron atlas covering all 135,168 Ouro-1.4B neurons, with per-loop top contexts and statistics.
- A pre-registered, paired steering test with permutation nulls built from random same-layer neurons, a multiplicity correction, a fluency guard, and power analysis on selection-free validation data.
- Measurements of loop structure: where neurons peak, and how similar their activations are across loops.
- Loop-resolved causal experiments: single-loop steering, test-time loop sweeps, and the saturation predicted by the sandwich norm.
- An interpretability audit of explanation transfer across loops.

Positive and negative results are both reported.

2 Related work

Looped and recurrent-depth language models. The models differ mainly in what they loop and how they inject the input:
- Huginn loops a middle block, with input injection and a randomly initialized latent state [1].
- Ouro loops the whole stack and learns an early-exit gate [2].
- Parcae stabilizes looping by constraining the spectral norm of the injection [3].

On the mechanistic side, Blayney et al. [5] find that recurrent blocks converge to cyclic fixed points, at which attention-head behavior stabilizes, and that they repeat feedforward-like stages of inference. Blog-length studies intervene on Ouro with residual-stream directions. One finds a last-loop steering advantage and apparent self-repair in Ouro-1.4B-Thinking [6]. Another, mainly about Nanbeige4.2-3B, finds in a follow-up on Ouro-2.6B that per-pass refusal directions converge from the second pass onward and that steering along the refusal direction removes refusal [7]. A study of a frozen 2.6B looped model (Ouro-RLTT) finds that internal readouts give validated decision-level gains, but that directional steering is negative and generative control yields no gain [9]. Self-repair after ablations is also known in feedforward models [10].

Neuron-level interpretability. There are three strands relevant here:
- Explaining neurons from max-activating examples, possibly with an LLM: [11], Delphi [12], which also introduced the detection scorer we use, and Transluce's descriptions of every Llama-3.1-8B-Instruct MLP neuron [13]. Output-centric descriptions, which describe a feature by its effect on outputs (vocabulary projection, or the tokens promoted when the feature is stimulated), complement activation-based ones [14].
- Studies of specific neuron families: universal [15], confidence-regulation [16], dead/n-gram/positional [17], sparse-probing [18], vocabulary-promoting value vectors [19], and gated (SwiGLU) neurons analysed through their input-output weight interaction [20].
- Goodfire's essay [8], which argues that most neuron poles of modern models have one main job. It reads, steers and ablates raw neurons, and compares ablations to random neurons from the

same layers. We adopt that design, extend the random same-layer control to steering, and make it quantitative with pre-registered tests.

Steering evaluation. Activation addition [21] is the basic residual-stream steering method. AxBench [22] rates steered text with an LLM judge for concept presence, instruction relevance and fluency (0–2 each); we use its concept and fluency prompts as a secondary metric. Our primary metric is LLM-free, so explanations cannot leak into their own evaluation.

Methodology. We follow guidance on avoiding ML pitfalls [23] (keep test data out of model selection; report statistical significance carefully) and the NeurIPS paper checklist [24] (error bars and statistical significance). Pre-registration, multiplicity control and reporting negative results are our own additions.

3 Method

3.1 Model and neuron definition

We study Ouro-1.4B [2] at revision `574fa66`. It has 24 decoder layers (d = 2048, 16-head MHA, SwiGLU MLP of width 5632, vocabulary 49,152). The whole stack is applied $T = 4$ times with shared weights. Each layer uses sandwich RMSNorm, and the residual stream is RMS-normalized after every loop. Ouro always executes all T loops; we set `early_exit_step = T-1`, so the logits always come from the last loop. The default early-exit path, which thresholds a cumulative exit distribution, disagreed with the fixed last-loop exit on 0 of 2,048 held-out positions (8 chunks × 256), but fixing the exit removes the dependence by construction. A **neuron** (ℓ, i) is element i of the SwiGLU activation entering `down_proj` in layer ℓ:

$$a_{\ell,i,t} = \mathrm{silu}(g_i^\top x) \cdot (u_i^\top x)$$

Here x is the post-attention-normalized residual at loop t. There are $24 \times 5632 = 135{,}168$ neurons, and each fires once per loop. Activations are signed, so following [8] we treat the positive and negative **poles** of every neuron as separate units. Reading and editing use one forward pre-hook per `down_proj`; the loop index is the per-layer call count modulo T. Unit tests check the hook against a manual SwiGLU recomputation at every loop, confirm that identity edits leave the logits bit-identical, and confirm that cached decoding matches full forward passes (same next token, probabilities within 0.02).

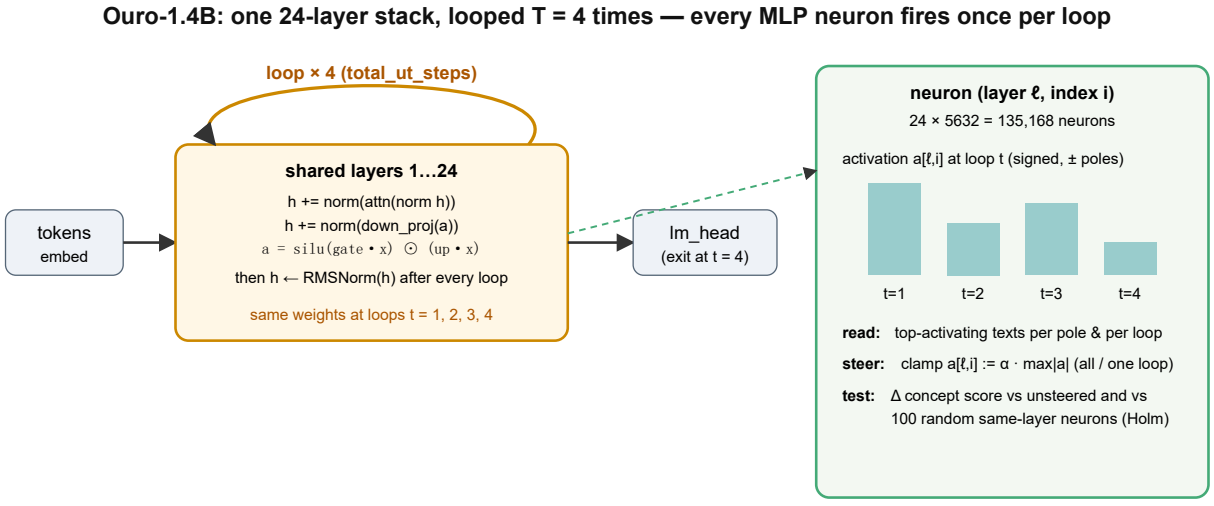


Figure 1: Object of study. Ouro-1.4B applies one 24-layer stack four times; every SwiGLU neuron is read and steered at the input of `down_proj`, once per loop, through a forward pre-hook.

3.2 Neuron atlas

We stream 8M tokens of documents drawn from FineWeb-Edu [25] with probability 0.9 and from OpenWebMath [26] with probability 0.1 (token share not measured), packed into 256-token chunks.

For every neuron and pole we keep the top-40 contexts, taking the maximum over positions and loops and recording the loop at which it occurs. We also keep the top-20 contexts at each loop, and per-loop moments and exceedance counts (computed on every 8th batch). The first 4 positions of each chunk are excluded.

3.3 Candidate selection on discovery data

Prompts are 12-token stems of WikiText-103 **test** paragraphs [27]; none of the 750 stems occurs verbatim in the 8M atlas tokens. They are split once, with a fixed seed, into discovery (150), confirmatory (300) and reserve (300) sets.

Selection. From a random pool of 4,000 alive neurons (8,000 units), we embed each unit's top-20 contexts with bge-small-en-v1.5 [28] and rank units by coherence (mean pairwise cosine). Units whose top contexts share a single peak token at least 50% of the time are excluded as token detectors [17]. The 200 most coherent units are clamped at $\alpha \in \{1, 2, 4\}$ times their maximum activation, on 25 discovery prompts, to obtain a quick estimate of the effect. The ten best units that pass a perplexity guard are then **validated** on 100 different discovery prompts, so that the effect sizes entering the power analysis are not inflated by selection. A unit is eligible if its validation mean $\Delta C > 0$, its median perplexity ratio is at most 2, and its median steered distinct-2 is at least 0.6 (a degeneracy guard; see Section 4.2). The five eligible units with the largest validation d_z are pre-registered.

3.4 Steering intervention and primary metric

Steering clamps $a_{\ell,i,t} \coloneqq c = \pm\alpha \cdot \max|a|$ (sign of the pole) at every position and, unless stated otherwise, every loop. Because the MLP write is RMS-normalized after `down_proj`, a single clamped neuron's contribution per layer and loop is bounded. As $|c| \to \infty$ the write tends to $\gamma \odot \operatorname{sign}(c) \frac{w_i}{\mathrm{rms}(w_i)}$, so the dose-response is predicted to saturate. The crossover scale is $c^* \approx \|r\| \,/\, \|w_i\|$, where r is the write of the other neurons; we measure it as the median of $\|W_{\text{down}}a\| \,/\, \|w_i\|$ over unsteered tokens.

For each prompt we sample $k = 3$ continuations of 48 tokens (temperature 0.7, top-p 0.9). Steered and unsteered runs share seeds and batch layout, so each steered sample is paired with an unsteered sample. (The random-neuron null runs of Test 2 reuse the seed in a different batch layout, so their pairing is not guaranteed bit-exact; unpaired noise can only widen the null.)

The **concept score** of a continuation is the cosine between its bge-small embedding and the unit's **contrastive concept direction** $u = \text{normalize}(\overline{e}_{\text{top}} - \overline{e}_{\text{random}})$. Here $\overline{e}_{\text{top}}$ is the mean embedding of the unit's top-20 atlas contexts and $\overline{e}_{\text{random}}$ the mean over 2,000 random atlas contexts. The primary per-prompt statistic is ΔC, the steered minus unsteered score averaged over the k samples. It involves no LLM, so a label cannot leak into its own evaluation.

3.5 Pre-registered confirmatory test

Candidates, α, n, the tests and the decision rule were written to `reports/preregistration.md` and committed before any confirmatory prompt was used. The pipeline refuses to run the test otherwise.

- **Test 1:** one-sided Wilcoxon signed-rank on per-prompt ΔC.
- **Test 2:** a random-neuron null, adapting the random same-layer **ablation** control of [8] to steering. $R = 100$ alive neurons from the candidate's layer are clamped at the same α times their own maximum on the same pole. The statistic is mean ΔC on the first 50 prompts (sample 0), scored with the **candidate's** concept direction, giving the empirical $p = (1 + \#\{\text{null} \geq \text{obs}\})/(R + 1)$ [29].
- **Multiplicity:** Holm [30] across the K candidates, separately for each test. $K \leq 5$ is forced because the smallest attainable permutation p, $1/101$, must be at most $0.05/K$.
- **Decision rule:** a neuron is a significant steering neuron iff both Holm-adjusted $p < 0.05$, the median perplexity ratio of steered to unsteered continuations (both scored by the unsteered model) is at most 2, and the median distinct-2 of steered continuations is at least 0.6.
- **Effect sizes:** mean ΔC with a percentile-bootstrap 95% CI (10,000 resamples); d_z and matched-pairs rank-biserial r [31] as point estimates.

- **Power:** n is chosen from the validation d_z by power analysis (one-sided paired t at $0.05/K$, power 0.9, divided by the Wilcoxon asymptotic relative efficiency $3/\pi \approx 0.955$ [32]), clipped to [100, 300].

3.6 Secondary analyses (post hoc unless stated)

These analyses are not part of the decision rule:

- an all-MiniLM-L6-v2 concept score;
- distinct-2 per continuation, a per-text variant of distinct-n [33];
- AxBench-style blind LLM ratings [22] from a local Qwen3.5-9B [34] served by llama.cpp [35];
- teacher-forced shift of next-token probability on each neuron's keyword set (up to 20 tokens over-represented near its atlas peaks), and three further sampling seed sets;
- dose-response over α;
- zero-ablation of the neuron on its held-out top contexts versus 100 random same-layer ablations;
- **loop locus** (clamping at a single loop);
- **test-time loops** ($T = 1..8$).

The loop structure of all neurons is summarized by:

- the loop holding each neuron's strongest context;
- the Jaccard overlap of top-20 contexts between loops;
- per-neuron Pearson correlation of activations between loops on the same held-out tokens.

The interpretability audit asks the local LLM, using the Delphi explainer system prompt (without its few-shot demonstrations) and the Delphi detection scorer [12], to describe 150 neurons separately at loop 1 and loop 4, and scores each explanation on held-out contexts from the same and from the other loop. Same-loop vs cross-loop and same-loop vs shuffled-explanation scores are compared with paired two-sided Wilcoxon tests.

4 Results

4.1 Model sanity and the neuron atlas

Held-out loss falls from 4.03 nats at $T = 1$ to 2.51 at the trained $T = 4$ (perplexity 12.3), and rises slightly when extrapolating to $T = 5..8$ (2.55–2.56). This parallels the benchmark trend reported for Ouro, which improves up to the trained depth and degrades moderately beyond it [2].

- **Atlas.** 8M tokens, 72.5 min. No neuron is dead. Median $\max|a|$ is 0.64.
- **Peak location.** Strongest activations concentrate in early loops: the share of neurons whose top context occurs at loop 1, 2, 3, 4 is 41.0%, 29.7%, 16.7%, 12.7% (both poles pooled).
- **Cross-loop similarity.** On 32,256 held-out tokens, the median per-neuron Pearson correlation between loop-1 and loop-4 activations is 0.54 (IQR 0.43–0.63). It rises with depth from ≈ 0 in layer 0 to 0.59–0.65 in layers 14–23.
- **Convergence.** Adjacent loops converge: median $r(1,2) = 0.65$, $r(2,3) = 0.84$, $r(3,4) = 0.94$, consistent with fixed-point dynamics [5].
- **Top-context overlap** between loop 1 and loop 4 is much lower (Jaccard 0.01–0.19 by layer, highest at layer 19; positive pole, 500 random neurons per layer). Top-20 sets cover only the extreme tail of the activations and are sensitive to near-ties; how much of the gap to the correlations this explains was not tested (Figure 2).

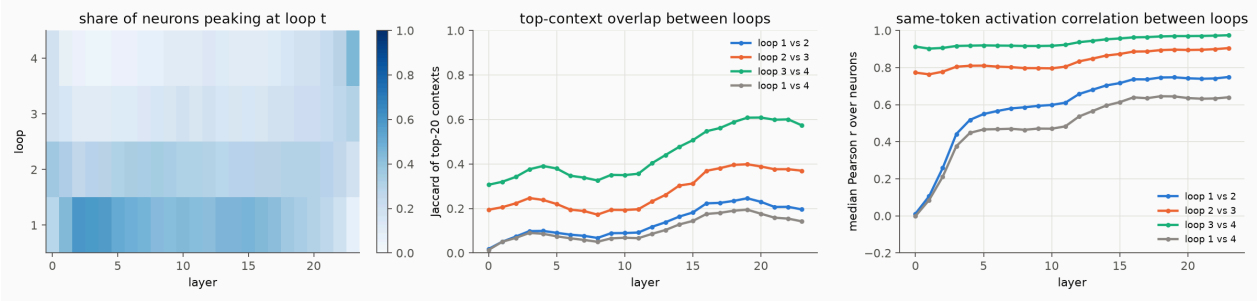


Figure 2: Loop structure by layer. Left: share of neuron poles (all 135,168 neurons, both poles) whose strongest atlas context occurs at each loop. Middle: Jaccard overlap of top-20 context chunks between loops (positive pole, 500 random alive neurons per layer). Right: median same-token activation correlation between loops over all neurons (32,256 held-out tokens).

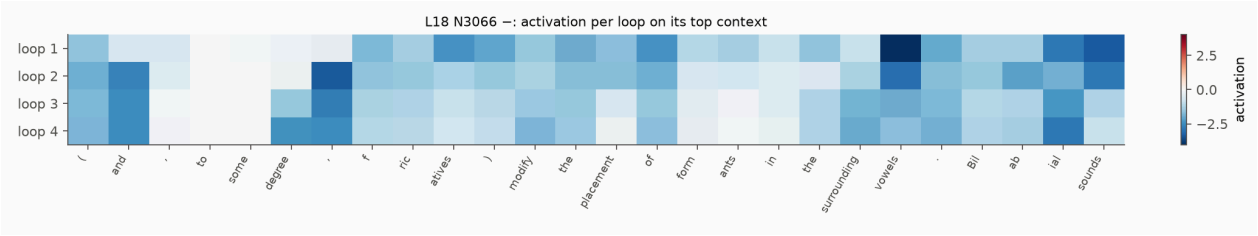


Figure 3: Per-loop activations of the phonetics neuron L18 N3066 (negative pole) on its strongest atlas context. The peak on "vowels" is largest at loop 1 and weakens at later loops, while other tokens gain activation, illustrating partial reuse of one neuron across loops.

4.2 Selection, and a failure mode of the fluency guard

Of 8,000 neuron poles, 21.2% are token detectors. The 200 most coherent of the remaining units were quick-steered.

The two units with the largest validated concept effects, an if-statement neuron (L19 N3901−, $d_z = 2.52$) and a neuron whose label mentions Fibonacci sequences among other math tokens (L0 N2848+, $d_z = 2.33$), turned out to reach them by degenerating into "ififf…" and "Fib Fib Fib…". This degenerate text has **lower** perplexity under the unsteered model than normal continuations, so a perplexity guard alone accepts it. We added a distinct-2 guard (steered median ≥ 0.6; unsteered text has ≈ 0.95 in the discovery validation run) before any confirmatory prompt was used, and recorded the amendment in the pre-registration.

4.3 Confirmatory test (H1, pre-registered)

Table 1 reports the pre-registered test.

All five candidates meet the decision rule, so H1 is supported.

- Wilcoxon $p \leq 1.1 \times 10^{-15}$ before correction for every neuron.
- Every bootstrap 95% CI of mean ΔC excludes zero.
- Each neuron's 50-prompt statistic exceeds all 100 (four neurons) or 99 of 100 (L4 N2104) random same-layer neurons clamped the same way. The candidates sit 7.3–12.1 null standard deviations (population SD) above the null mean.
- The Holm-adjusted permutation $p = 0.0495$ is the smallest value attainable with $K = 5$ and $R = 100$. It reflects the resolution of the null, not a marginal effect.
- Effects are large: $d_z = 0.99$–1.61 and matched-pairs rank-biserial $r = 0.91$–0.99.
- The secondary MiniLM embedder agrees ($\Delta C = 0.108$–0.248).
- Steering costs fluency but stays within the guard: median perplexity ratio 1.58–1.83, and steered distinct-2 is 0.80–0.91 (unsteered text had ≈ 0.95 in the discovery validation run).

Neuron	α	Concept (top contexts)	mean ΔC [95% CI]	d_z	Holm p_{Wil}	p_{null} (Holm)	z vs null	PPL ratio
L18 N3066 −	1	phonetics, letters	0.179 [0.155, 0.204]	1.42	2.4e-17	0.0099 (0.0495)	12.1	1.58
L13 N1680 +	1	non-duality, dependent origination	0.141 [0.119, 0.163]	1.26	1.7e-16	0.0099 (0.0495)	11.3	1.62
L4 N2104 +	2	pedagogy	0.130 [0.112, 0.148]	1.43	2.5e-17	0.0198 (0.0495)	7.6	1.65
L6 N3344 −	4	forum “mark as accepted answer”	0.167 [0.147, 0.188]	1.61	2.4e-17	0.0099 (0.0495)	7.3	1.83
L18 N1735 −	2	electron orbitals, bonding	0.127 [0.103, 0.153]	0.99	1.1e-15	0.0099 (0.0495)	11.2	1.72

Table 1: Pre-registered confirmatory test on 100 held-out prompts × 3 paired samples. ΔC: change in concept score (bge-small). p_{null}: empirical p against 100 random same-layer neurons. z vs null is descriptive (post hoc). All five rows meet the decision rule. Source: `results/Ouro-1.4B/confirm.json`.

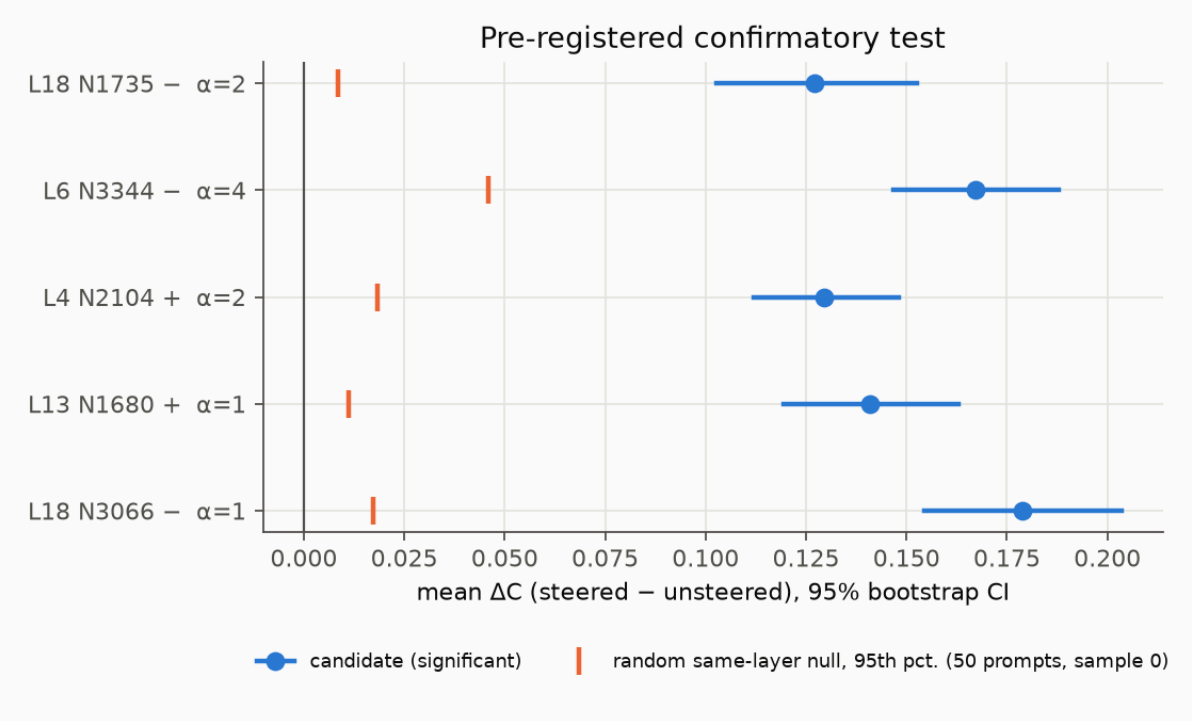


Figure 4: Mean ΔC (100 prompts × 3 samples) with 95% bootstrap CIs (blue) and the 95th percentile of the random same-layer neuron null (orange tick; the null statistic uses the first 50 prompts, sample 0) for each candidate.

Steered text visibly carries the concepts. The examples below are the second confirmatory prompt, sample 0 (chosen from the first three prompts for visible concept content; the eight stored prompts per neuron are in `confirm_examples.jsonl`):

- L18 N3066− (phonetics) continues “In the Philippines, Nepartak produced strong winds, heavy” with “rains, and severe flooding. The sounds of the letters /p/ and /t/ are softened. In the International Phonetic Alphabet (IPA) …”.
- The unsteered model continues the same prompt with casualty and damage figures.
- L4 N2104+ (pedagogy) instead writes “rains, and clear skies. - Aarav, a Hindu sage, taught his students through a unique method.”, then a heading “The Role of an Educator” and “An educator, a dedicated instructor, plays a pivotal …”.

4.4 Follow-up experiments (post hoc)

These analyses use 60 held-out prompts, 1 sample each.

Specificity. Steering neuron i and scoring concept j gives a matrix whose diagonal is the largest entry of each row, 1.4–3.7× the largest off-diagonal entry (Figure 5). The smallest margin pairs the forum “accepted answer” neuron with the programming concept, an overlap that is plausible for Q&A text.

Dose-response. ΔC rises with α and then levels off for four of six neurons, qualitatively as the sandwich-norm bound predicts; no curve was fitted (dips of at most 0.011, inside the CIs, make the rise not strictly monotone). Neurons with larger $c^*/\max|a|$ (2.2–2.6) are still rising at $\alpha = 16$; those near 1 plateau from $\alpha \approx 2$–8. Clamping to the opposite sign does not steer toward the concept ($|\Delta C| \leq 0.04$; four of twelve conditions have a small negative effect whose CI excludes 0), consistent with each concept being carried by one pole. Perplexity ratio exceeds 2 for most neurons at $\alpha \geq 4$.

Loop locus (H2). Clamping a single loop barely steers: $\Delta C \leq 0.043$, against 0.13–0.17 for all four loops, for the five confirmed neurons. The four single-loop effects sum to −2% to 30% of the all-loop effect for four of them (76% for L13 N1680).

- Steering in a looped model is therefore strongly super-additive across loops. One injection is largely undone by the following loops, consistent with self-repair (suggested for Ouro by [6]; shown for feedforward models by [10]).
- The exploratory if-statement neuron is the exception. Its last loop alone gives 56% of the full effect ($\Delta C = 0.176$, $p = 1.7 \times 10^{-10}$), while loops 1–3 give nothing, matching the last-loop advantage reported for residual-direction steering of Ouro-1.4B-Thinking [6].
- Clamping at loop 4 moves the final hidden state most (relative displacement 0.20–0.47, against 0.12–0.23 at loops 1–3).

Test-time loops (H3). Clamping at every loop while running $T = 1..8$:

- the concept effect is already present at $T = 1$ (ΔC 0.10–0.35);
- it does not grow systematically with T: for the confirmed neurons, ΔC at $T = 1$ is 70–99% of its $T = 4$ value, and from $T = 2$ to 8 it fluctuates between 0.11 and 0.25 without a monotone trend;
- next-token KL grows with T, apart from small dips (≤ 0.06) for three of six neurons.

More loops add more injections and more distribution shift, but not proportionally more concept.

Necessity. Zero-ablating a neuron on its held-out top contexts increases the next-token loss beyond all 100 random same-layer ablations for 4 of 6 neurons (unadjusted $p = 0.0099$; $z = 3.1$–53.6). This is exploratory; correcting with Holm across six would give 0.059. Two confirmed neurons are sufficient for steering but not necessary for prediction, and ablating L6 N3344 even **improves** prediction ($z = -8.9$).

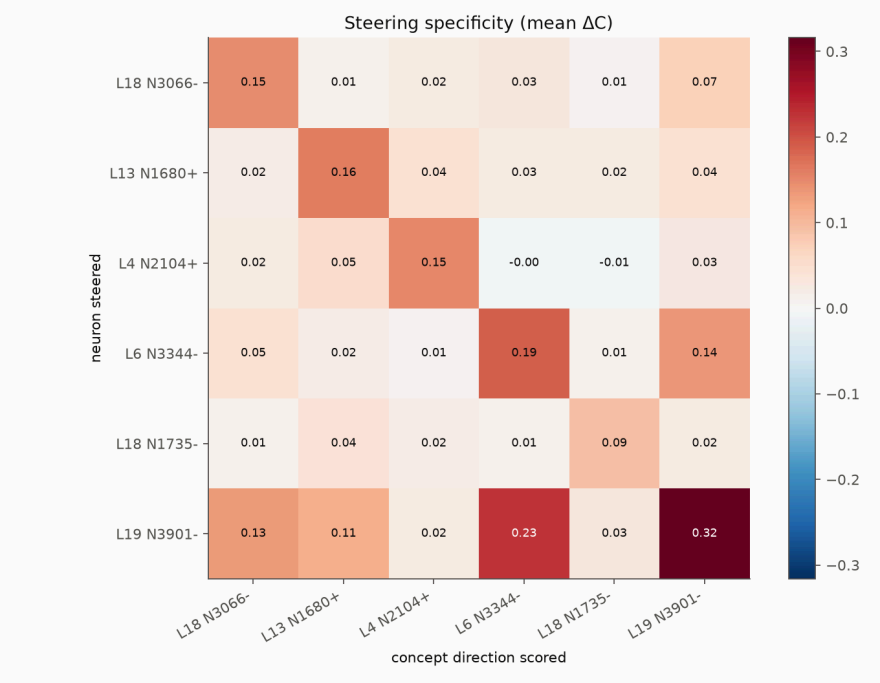


Figure 5: Steering specificity: mean ΔC on each concept (columns) when steering each neuron (rows). Diagonal entries dominate their rows.

4.5 Interpretability audit (H4)

Delphi explanations written by the local Qwen3.5-9B from 12 examples per loop detect held-out activating contexts above chance (loop-4 explanations: balanced accuracy 0.60 vs 0.50 for shuffled explanations on the same contexts, paired difference +0.097, two-sided Wilcoxon $p = 1.4 \times 10^{-7}$, $n = 123$). However, they transfer across loops as well as within a loop (same 0.60, cross 0.59; paired difference +0.012, two-sided Wilcoxon $p = 0.20$, $n = 240$).

H4 (loop-specific explanations) is therefore not supported. Only 25% (loop 1, Wilson 95% CI 18–33%) and 21% (loop 4, CI 15–28%) of neurons reach balanced accuracy ≥ 0.75, so the weak explainer limits how sensitive this test can be, and the activation correlations above are the more sensitive evidence.

4.6 Robustness and mechanism (plan-completion analyses, post hoc)

- **Seed sensitivity.** Re-running the confirmatory comparison with three new seed sets (100 prompts, 1 sample each) gives $\Delta C = 0.11$–0.18 for every neuron in every set ($p \leq 2.9 \times 10^{-8}$). The spread across seed sets is 0.007–0.024.
- **Keyword-probability shift.** Teacher-forced on unsteered continuations, clamping a neuron raises the next-token probability mass on its keyword set by 0.45–0.94 nats on average (≈ 1.6–$2.6\times$, all $p \approx 2 \times 10^{-18}$). The keywords are up to 20 tokens over-represented around its atlas peaks, e.g. "sounds, vowel" for phonetics, "pedagogy, instruction" for pedagogy, "orbital, shell, outermost" for orbitals. The lists also contain function words and punctuation (the non-duality neuron's list is almost only function words), so this is a peak-context vocabulary rather than a curated concept vocabulary.
- **Self-repair ratio.** A clamp at loop 1, 2 or 3 leaves 33–78% of its immediate residual displacement at the output (six neurons, including the exploratory one), while a clamp at loop 4 leaves 67–126% (above 100% for four of six). Later loops thus shrink earlier injections, quantifying the super-additivity above.
- **Silencing.** Holding a neuron at 0 during generation does not change the concept score ($|\Delta C| \leq 0.005$).
- **Saturation in T.** Next-token KL saturates with the number of loops ($\tau \approx 2$–4 loops, $R^2 = 0.90$–0.99, five confirmed neurons), while the concept effect is already near its plateau at $T = 1$–2.

- **Positional neurons.** None of the 10 validated candidates is positional (no position holds more than 3 of the 40 top peaks, i.e. 7.5%).

4.7 Replication on Ouro-1.4B-Thinking

The five neuron indices were tested unchanged in Ouro-1.4B-Thinking, which is Ouro-1.4B after supervised fine-tuning on about 8.3M reasoning examples [2]. There was no re-selection.

Reasoning SFT barely moves the MLP neurons:

- the per-neuron cosine between base and Thinking weights (minimum over gate row, up row and down column) is ≥ 0.993 for 99% of neurons in the four candidate layers, and ≥ 0.996 for the five candidates;
- the candidates' maximum activations change by at most 5.1% of the base value.

All five neurons steer the post-trained model with similar or larger effects:

- $\Delta C = 0.161$–0.180, against 0.127–0.179 in the base model (larger for four neurons; for L18 N3066 it is 0.171 against 0.179, within the confidence intervals);
- Holm-adjusted Wilcoxon $p \leq 2.1 \times 10^{-16}$;
- each neuron exceeds all 20 random same-layer neurons;
- the PPL ratio is at most 1.98.

With $R = 20$ the smallest permutation p is $1/21$, which cannot pass Holm at $K = 5$, so the random-neuron comparison is descriptive here. The code at the start of the run named the Wilcoxon test as the replication criterion; the "beats all 20" and guard conditions were written down after the run.

A second replication on 100 never-used prompts (the reserve split) gives the same picture:

- $\Delta C = 0.157$–0.179;
- Holm-adjusted Wilcoxon $p \leq 2.0 \times 10^{-16}$;
- each neuron again beats all 20 random neurons;
- the PPL ratio is at most 1.97.

4.8 LLM judge (secondary, pre-registered)

A local Qwen3.5-9B rated continuations of 50 confirm prompts using the AxBench concept and fluency templates [22]. Each continuation was rated by itself, in a shuffled order, so the judge was blind to condition.

Concept presence (0–2). The neuron's Qwen label served as the concept, and ratings rose from 0–0.02 unsteered to:

- 0.26 for L18 N3066−;
- 0.24 for L13 N1680+;
- 0.26 for L4 N2104+;
- 0.36 for L18 N1735−;
- 0.06 for L6 N3344−.

Paired one-sided Wilcoxon $p = 0.003$, 0.0007, 0.0007, 0.0001 and 0.042 respectively. All remain below 0.05 after Holm (largest Holm-adjusted $p = 0.042$; `reports/tables/audit_judge.md`).

Fluency (0–2). Fluency fell from 1.50 to 0.90–1.17 (95% CIs of the paired differences exclude 0), which agrees with the perplexity cost.

The judge is strict: absolute concept ratings are low, and the forum-answer neuron (L6 N3344), whose concept is the most abstract, is the weakest by this measure. The ranking of neurons only partly matches the embedding metric.

5 Discussion

Single neurons are effective steering handles in a looped LM. Goodfire's essay argues that raw neurons are often clean, single-purpose units [8]. Our results support a narrower causal version of that

claim for a looped model. Neurons chosen for coherent top contexts include units that steer generation toward their concept, and they do so far beyond random neurons from the same layer clamped the same way. The effect survives a change of embedding model and reasoning post-training, and it is specific to each neuron's own concept.

Looping makes steering cumulative. A clamp applied at one loop is largely undone by the remaining loops. Only repeated injection, at every pass through the shared weights, changes behavior. Steering a neuron at all four loops works well, while for four of the five confirmed neurons any single loop is nearly inert (for L13 N1680, single-loop effects are small but non-zero). Mechanistically, an injection at loops 1–3 keeps only 33–78% of its size by the time it reaches the output. Behaviorally this matches the last-loop advantage reported for residual directions in Ouro-1.4B-Thinking [6], but the mechanism differs: there, early perturbations persist and are rotated away from the concept direction, while here the injected displacement itself shrinks. It is also consistent with the convergence of activations across loops (median same-token activation correlation between consecutive loops rises from 0.65 to 0.94).

The exception, the if-statement neuron, acts mainly at the last loop. That suggests the location of the effect depends on the neuron: some concepts are written late, close to the output, while others are maintained throughout.

The neuron's function is mostly, but not entirely, shared across loops. Same-token activations at loop 1 and loop 4 correlate at a median of 0.54. Neurons are therefore partly reused for different inputs at different depths, as the input distribution of every layer shifts from loop to loop. Explanations written by a local 9B model do not detect this difference (same-loop 0.60 vs cross-loop 0.59). With explainers this weak, activation statistics are the more sensitive instrument.

Fluency guards must catch repetition. Concept scores can be maximized by degenerate repetition, which a language model finds **easy** to predict. Perplexity-based fluency guards can therefore pass degenerate outputs. A lexical-diversity guard, or an LLM fluency judge as in AxBench [22], is needed alongside them.

6 Limitations

- **Scope.** One looped model family (Ouro-1.4B and its Thinking variant) and 5 confirmed neurons, chosen for steerability from a pool of 200 coherent units. These are existence claims and say nothing about how common steerable neurons are. Ouro-2.6B (48 looped layers) was not attempted: its bf16 checkpoint is 5.3 GB [36] and its KV cache per token is twice Ouro-1.4B's, which leaves too little of the 8 GB GPU for the batched runs used here.
- **Metric.** The primary metric is an embedding-based concept score built from each neuron's own top contexts. It measures movement toward that contrastive direction, not human-judged concept presence. The LLM judge is a secondary check from a small local model.
- **Intervention.** We clamp the neuron at every position, including the prompt, and far beyond its natural activation pattern (α up to 4 for confirmation). Steering costs fluency (perplexity ratio 1.6–1.8).
- **Null resolution.** With $R = 100$ random neurons and $K = 5$, the permutation test's Holm-adjusted p cannot go below 0.0495. We report z-margins (7–12 SD) as descriptive support.
- **Post hoc analyses.** Follow-ups use 60 prompts and 1 sample. Ablation p-values are unadjusted. The T sweep beyond $T = 4$ is extrapolation.
- **Atlas size.** 8M tokens is far smaller than the 100M tokens used by Goodfire, so rare features are under-sampled.
- **Compute.** Everything ran on one 8 GB laptop GPU. The labeler is a 4-bit 9B model.

7 Conclusion

In Ouro-1.4B, a looped transformer that applies its 24 layers four times, we built a 135k-neuron atlas and ran a pre-registered, paired steering test with random same-layer neuron nulls. All five pre-registered

neurons (phonetics, non-duality, pedagogy, forum answers, electron orbitals) steer generation toward their concept with large, statistically significant effects. They are specific to their own concepts, level off with clamp strength for four of six neurons (qualitatively consistent with the sandwich-norm bound), and replicate in the reasoning-tuned Ouro-1.4B-Thinking.

Steering a neuron in a looped model requires clamping it at every loop. Single-loop interventions are mostly undone by the following loops. Neurons are largely reused across loops, but not identically: their same-token activation correlation between the first and last loop has a median of 0.54.

Bibliography

[1] J. Geiping *et al.*, "Scaling up Test-Time Compute with Latent Reasoning: A Recurrent Depth Approach," *arXiv preprint arXiv:2502.05171*, 2025, [Online]. Available: https://arxiv.org/abs/2502.05171

[2] R.-J. Zhu *et al.*, "Scaling Latent Reasoning via Looped Language Models," *arXiv preprint arXiv:2510.25741*, 2025, [Online]. Available: https://arxiv.org/abs/2510.25741

[3] H. Prairie, Z. Novack, T. Berg-Kirkpatrick, and D. Y. Fu, "Parcae: Scaling Laws For Stable Looped Language Models," *arXiv preprint arXiv:2604.12946*, 2026, [Online]. Available: https://arxiv.org/abs/2604.12946

[4] ByteDance Seed, "Ouro-1.4B model card." [Online]. Available: https://huggingface.co/ByteDance/Ouro-1.4B

[5] H. Blayney *et al.*, "A Mechanistic Analysis of Looped Reasoning Language Models," *arXiv preprint arXiv:2604.11791*, 2026, [Online]. Available: https://arxiv.org/abs/2604.11791

[6] nesiacel, "A recurrent LLM is quite easy to interpret but complex to steer." [Online]. Available: https://www.lesswrong.com/posts/rzcJMpnehFBD686gj/a-recurrent-llm-is-quite-easy-to-interpret-but-complex-to

[7] M. Suiche, "Steering a Loop: Control Vectors Meet the Looped Transformer." [Online]. Available: https://msuiche.com/posts/abliterating-a-loop-control-vectors-meet-the-looped-transformer/

[8] N. Cammarata, "Neural Networks Are Already Clean." [Online]. Available: https://already-clean.goodfire.com/

[9] J. Kirin, "Operational Proto-Introspection in Looped Language Models: Process-Quality Taps, Executable Branching, and the Readout-Control Boundary," *arXiv preprint arXiv:2607.18553*, 2026, [Online]. Available: https://arxiv.org/abs/2607.18553

[10] T. McGrath, M. Rahtz, J. Kramár, V. Mikulik, and S. Legg, "The Hydra Effect: Emergent Self-repair in Language Model Computations," *arXiv preprint arXiv:2307.15771*, 2023, [Online]. Available: https://arxiv.org/abs/2307.15771

[11] S. Bills *et al.*, "Language models can explain neurons in language models." [Online]. Available: https://openaipublic.blob.core.windows.net/neuron-explainer/paper/index.html

[12] G. Paulo, A. Mallen, C. Juang, and N. Belrose, "Automatically Interpreting Millions of Features in Large Language Models," *arXiv preprint arXiv:2410.13928*, 2024, [Online]. Available: https://arxiv.org/abs/2410.13928

[13] D. Choi, V. Huang, K. Meng, D. D. Johnson, J. Steinhardt, and S. Schwettmann, "Scaling Automatic Neuron Description." [Online]. Available: https://transluce.org/neuron-descriptions

[14] Y. Gur-Arieh, R. Mayan, C. Agassy, A. Geiger, and M. Geva, "Enhancing Automated Interpretability with Output-Centric Feature Descriptions," *arXiv preprint arXiv:2501.08319*, 2025, [Online]. Available: https://arxiv.org/abs/2501.08319

[15] W. Gurnee *et al.*, "Universal Neurons in GPT2 Language Models," *arXiv preprint arXiv:2401.12181*, 2024, [Online]. Available: https://arxiv.org/abs/2401.12181

[16] A. Stolfo *et al.*, "Confidence Regulation Neurons in Language Models," *arXiv preprint arXiv:2406.16254*, 2024, [Online]. Available: https://arxiv.org/abs/2406.16254

[17] E. Voita, J. Ferrando, and C. Nalmpantis, "Neurons in Large Language Models: Dead, N-gram, Positional," *arXiv preprint arXiv:2309.04827*, 2023, [Online]. Available: https://arxiv.org/abs/2309.04827

[18] W. Gurnee, N. Nanda, M. Pauly, K. Harvey, D. Troitskii, and D. Bertsimas, "Finding Neurons in a Haystack: Case Studies with Sparse Probing," *arXiv preprint arXiv:2305.01610*, 2023, [Online]. Available: https://arxiv.org/abs/2305.01610

[19] M. Geva, A. Caciularu, K. R. Wang, and Y. Goldberg, "Transformer Feed-Forward Layers Build Predictions by Promoting Concepts in the Vocabulary Space," *arXiv preprint arXiv:2203.14680*, 2022, [Online]. Available: https://arxiv.org/abs/2203.14680

[20] S. Gerstner and H. Schütze, "Understanding Gated Neurons in Transformers from Their Input-Output Functionality," *arXiv preprint arXiv:2505.17936*, 2025, [Online]. Available: https://arxiv.org/abs/2505.17936

[21] A. M. Turner *et al.*, "Steering Language Models With Activation Engineering," *arXiv preprint arXiv:2308.10248*, 2023, [Online]. Available: https://arxiv.org/abs/2308.10248

[22] Z. Wu *et al.*, "AxBench: Steering LLMs? Even Simple Baselines Outperform Sparse Autoencoders," *arXiv preprint arXiv:2501.17148*, 2025, [Online]. Available: https://arxiv.org/abs/2501.17148

[23] M. A. Lones, "How to avoid machine learning pitfalls: a guide for academic researchers," *arXiv preprint arXiv:2108.02497*, 2021, [Online]. Available: https://arxiv.org/abs/2108.02497

[24] NeurIPS, "NeurIPS Paper Checklist Guidelines." [Online]. Available: https://neurips.cc/public/guides/PaperChecklist

[25] G. Penedo *et al.*, "The FineWeb Datasets: Decanting the Web for the Finest Text Data at Scale," *arXiv preprint arXiv:2406.17557*, 2024, [Online]. Available: https://arxiv.org/abs/2406.17557

[26] K. Paster, M. Dos Santos, Z. Azerbayev, and J. Ba, "OpenWebMath: An Open Dataset of High-Quality Mathematical Web Text," *arXiv preprint arXiv:2310.06786*, 2023, [Online]. Available: https://arxiv.org/abs/2310.06786

[27] S. Merity, C. Xiong, J. Bradbury, and R. Socher, "Pointer Sentinel Mixture Models," *arXiv preprint arXiv:1609.07843*, 2016, [Online]. Available: https://arxiv.org/abs/1609.07843

[28] S. Xiao, Z. Liu, P. Zhang, N. Muennighoff, D. Lian, and J.-Y. Nie, "C-Pack: Packed Resources For General Chinese Embeddings," *arXiv preprint arXiv:2309.07597*, 2023, [Online]. Available: https://arxiv.org/abs/2309.07597

[29] B. Phipson and G. K. Smyth, "Permutation P-values Should Never Be Zero: Calculating Exact P-values When Permutations Are Randomly Drawn," *Statistical Applications in Genetics and Molecular Biology*, vol. 9, no. 1, 2010, [Online]. Available: https://arxiv.org/abs/1603.05766

[30] S. Holm, "A Simple Sequentially Rejective Multiple Test Procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979, [Online]. Available: https://www.jstor.org/stable/4615733

[31] D. S. Kerby, "The Simple Difference Formula: An Approach to Teaching Nonparametric Correlation," *Comprehensive Psychology*, vol. 3, 2014, [Online]. Available: https://doi.org/10.2466/11.IT.3.1

[32] J. L. Hodges and E. L. Lehmann, "The Efficiency of Some Nonparametric Competitors of the t-Test," *The Annals of Mathematical Statistics*, vol. 27, no. 2, pp. 324–335, 1956, [Online]. Available: https://doi.org/10.1214/aoms/1177728261

[33] J. Li, M. Galley, C. Brockett, J. Gao, and B. Dolan, "A Diversity-Promoting Objective Function for Neural Conversation Models," in *Proceedings of NAACL-HLT 2016*, 2016. [Online]. Available: https://arxiv.org/abs/1510.03055

[34] Qwen Team, "Qwen3.5." [Online]. Available: https://huggingface.co/Qwen/Qwen3.5-9B

[35] G. Gerganov and contributors, "llama.cpp." [Online]. Available: https://github.com/ggml-org/llama.cpp

[36] ByteDance Seed, "Ouro-2.6B model files." [Online]. Available: https://huggingface.co/ByteDance/Ouro-2.6B

AI statement. The codebase, experiments and this paper were written by Claude Code (Anthropic), an AI coding agent, which executed the exact specification provided by a human (Libor Burian): Goodfire-style neuron-level analysis of a looped transformer language model, aiming to find at least one neuron whose steering changes behavior with statistical significance. Human cleanup and rewriting of the code and text are in progress.